1

```
set ns [new Simulator]
set trf [open 1.tr w]
$ns trace-all $trf
set namf [open 1.nam w]
$ns namtrace-all $namf
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
$n0 label "source node"
$n1 label "Intermediate node"
$n2 label "destination node"
$ns duplex-link $n0 $n1 100Kb 100ms DropTail
$ns duplex-link $n1 $n2 5Mb 200ms DropTail
$ns queue-limit $n0 $n1 10
set tcp [new Agent/TCP]
$tcp set fid_ 1
$ns color 1 Blue
$ns attach-agent $n0 $tcp
set sink [new Agent/TCPSink]
$sink set fid_ 2
$ns color 2 Red
$ns attach-agent $n2 $sink
$ns connect $tcp $sink
set ftp [new Application/FTP]
$ftp attach-agent $tcp
$ftp set packetSize_ 500
$ftp set interval_ 0.001
```

```
proc finish {} {
   global ns trf namf
   $ns flush-trace
   close $trf
   close $namf
   exec nam 1.nam &
   exit 0
}
$ns at 0.1 "$ftp start"
$ns at 10.0 "finish"
$ns run
2
set ns [new Simulator]
set trf [open 2.tr w]
$ns trace-all $trf
set namf [open 2.nam w]
$ns namtrace-all $namf
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
set n3 [$ns node]
$n0 label "TCP Source"
$n1 label "UDP Source"
$n2 label "Router"
$n3 label "Destination"
$ns duplex-link $n0 $n2 10Mb 10ms DropTail
$ns duplex-link $n1 $n2 10Kb 100ms DropTail
$ns duplex-link $n2 $n3 10Kb 100ms DropTail
```

```
set tcp [new Agent/TCP]
$ns attach-agent $n0 $tcp
set sink [new Agent/TCPSink]
$ns attach-agent $n3 $sink
set ftp [new Application/FTP]
$ftp attach-agent $tcp
set udp [new Agent/UDP]
$ns attach-agent $n1 $udp
set null [new Agent/Null]
$ns attach-agent $n3 $null
set cbr [new Application/Traffic/CBR]
$cbr attach-agent $udp
$ns connect $tcp $sink
$ns connect $udp $null
$ftp set packetSize_ 200
$ftp set interval_ 0.001
$cbr set packetSize_ 300
$cbr set interval_ 0.001
$ns color 1 red
$ns color 2 green
$tcp set class_ 1
$udp set class_ 2
proc finish {} {
   global ns trf namf
   $ns flush-trace
   exec nam 2.nam &
   close $namf
   close $trf
```

```
    exit 0
}
$ns at 0.1 "$cbr start"
$ns at 0.3 "$ftp start"
$ns at 5.0 "$ftp stop"
$ns at 5.0 "$cbr stop"
$ns at 7.0 "finish"
$ns run
```

3

```
set ns [new Simulator]
set trf [open 3.tr w]
$ns trace-all $trf
set namf [open 3.nam w]
$ns namtrace-all $namf
se n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
set n3 [$ns node]
set n4 [$ns node]
set n5 [$ns node]
$ns duplex-link $n0 $n2 100Kb 10ms DropTail
$ns duplex-link $n1 $n2 100Kb 10ms DropTail
$ns duplex-link $n2 $n3 100Kb 10ms DropTail
$ns duplex-link $n3 $n4 100Kb 10ms DropTail
$ns duplex-link $n4 $n5 100Kb 10ms DropTail
set ping0 [new Agent/Ping]
$ns attach-agent $n0 $ping0
set ping2 [new Agent/Ping]
```

```
$ns attach-agent $n2 $ping2
$ns connect $ping0 $ping2
$ns color 1 red
$ns color 2 green
$ping0 set class_ 1
$ping2 set class_ 2
Agent/Ping instproc recv {from rtt} {
   $self instvar node_
   puts "The node [$node_ id] received a reply from $from with RTT = $rtt"
}
proc finish {} {
   global ns trf namf
   $ns flush-trace
   close $trf
   close $namf
   exec nam 3.nam &
   exit 0
}
$ns at 0.2 "$ping0 send"
$ns at 0.4 "$ping0 send"
$ns at 0.6 "$ping0 send"
$ns at 0.8 "$ping0 send"
$ns at 1.0 "finish"
$ns run
```

4

```
set ns [new Simulator]
set ntrace [open 4.tr w]
$ns trace-all $ntrace
```

```
set namfile [open 4.nam w]
$ns namtrace-all $namfile
$ns color 1 Blue
$ns color 2 Red
for {set i 0} {$i < 6} {incr i} {
   set n($i) [$ns node]
}
$ns duplex-link $n(0) $n(2) 2Mb 10ms DropTail
$ns duplex-link $n(1) $n(2) 2Mb 10ms DropTail
$ns simplex-link $n(2) $n(3) 0.3Mb 100ms DropTail
$ns simplex-link $n(3) $n(2) 0.3Mb 100ms DropTail
set lan [$ns newLan "$n(3) $n(4) $n(5)" 0.5Mb 40ms LL Queue/DropTail MAC/802_3 Channel]
$ns duplex-link-op $n(0) $n(2) orient right-down
$ns duplex-link-op $n(1) $n(2) orient right-up
$ns simplex-link-op $n(2) $n(3) orient right
$ns queue-limit $n(2) $n(3) 10
$ns simplex-link-op $n(2) $n(3) queuePos 0.5
set tcp [new Agent/TCP]
$tcp set fid_ 1
$tcp set packetSize_ 552
$ns attach-agent $n(0) $tcp
set sink [new Agent/TCPSink]
$ns attach-agent $n(4) $sink
$ns connect $tcp $sink
set ftp [new Application/FTP]
$ftp attach-agent $tcp
set udp [new Agent/UDP]
```

```
$udp set fid_ 2
$ns attach-agent $n(1) $udp
set null [new Agent/Null]
$ns attach-agent $n(5) $null
$ns connect $udp $null
set cbr [new Application/Traffic/CBR]
$cbr set packetSize_ 1000
$cbr set interval_ 0.01
$cbr attach-agent $udp
proc finish {} {
   global ns ntrace namfile
   $ns flush-trace
   close $ntrace
   close $namfile
   exec nam 4.nam &
   set tcpsize [exec grep "^r " 4.tr | grep "tcp" | cut -d " " -f 6 | tail -n 1]
   set numTcp [exec grep "^r" 4.tr | grep -c "tcp"]
   set tcpTime 23.0
   puts "the throughput of ftp"
   puts "[expr ($numTcp * $tcpsize) / $tcpTime] bytes per second"
   set udpsize [exec grep "^r " 4.tr | grep "cbr" | cut -d " " -f 6 | tail -n 1]
   set numudp [exec grep "^r" 4.tr | grep -c "cbr"]
   set udpTime 24.0
   puts "the throughput of cbr"
   puts "[expr ($numudp * $udpsize) / $udpTime] bytes per second"
}
$ns at 0.1 "$cbr start"
$ns at 1.0 "$ftp start"
```

```
$ns at 24.0 "$ftp stop"

$ns at 24.5 "$cbr stop"

$ns at 25.0 "finish"

$ns run
```